Digital Communication IC Design

Lecture 1: Fourier Transform and Random Process

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Communication Basics

- ❑ Fourier Series and Fourier Transform
- ❑ Random Process
- ❑ Sampling Theory
- ❑ Discrete Fourier Transform

Fourier Series

- A periodic signal (waveform) $f(t)$ represented by Fourier series

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n cos(nwt) + \sum_{n=1}^{\infty} b_n sin(nwt)$$

  - Angular frequency (the fundamental frequency or the first harmonic related to the fundamental frequency f) $w = 2\pi/T_p$;
  - $f = 1/T_p$, hence, $w = 2\pi f$
  - T_p: the period of the signal (or waveform)
  - a_0 is a constant equal to the time average of $f(t)$ over one period. It may represent a **dc** voltage level.

$$a_0 = \frac{1}{T_p} \int_{-T_p/2}^{T_p/2} f(t) dt$$

Fourier Series (cont'd)

$$a_n = \frac{2}{T_p} \int_{-T_p/2}^{T_p/2} f(t) cos(nwt) dt$$

and

$$b_n = \frac{2}{T_p} \int_{-T_p/2}^{T_p/2} f(t) sin(nwt) dt$$

- a_n : the amplitudes of cos-sinusoidal frequency-dependent terms at the positive harmonic frequencies nw $(= 2\pi nf)$
- b_n : the amplitudes of sinusoidal frequency-dependent terms at the positive harmonic frequencies $nw(= 2\pi nf)$
- nw: the nth harmonics of w

Fourier Series (cont'd)

- Using exponential notation to represent Fourier series

$$f(t) = \sum_{n=-\infty}^{\infty} d_n e^{jnwt}$$

and

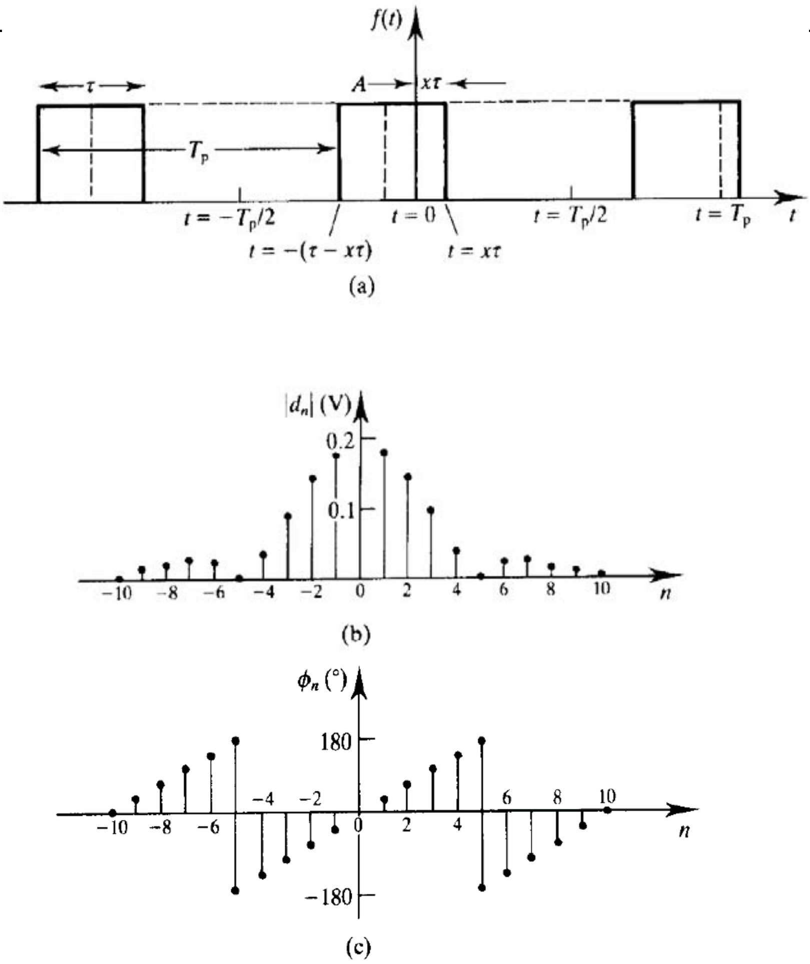
$$d_n = \frac{1}{T_p}\int_{-T_p/2}^{T_p/2} f(t)e^{-jnwt}dt$$

  - d_n is complex and $|d_n|$ has the unit of volts.
  - The complex and trigonometric forms

$$d_n = \sqrt{a_n^2 + b_n^2}$$

and

$$\phi_n = tan^{-1}(b_n/a_n)$$



Fourier Transform

- Aperiodic signal (waveform) expressed by Fourier transform
- A most important example of a single rectangular pulse
  - $T_p \to \infty$ (increase the spacing between the harmonic components)
  - $1/T_p = w/2\pi \to dw/2\pi \approx 0$
  - Change the discrete frequency variable $\boldsymbol{nw}$ to the continuous variable $\boldsymbol{w}$, and the amplitude and phase spectra become continuous ($d_n \to d(w)$ and $T_p \to \infty$)

Fourier Transform (cont'd)

- Fourier transform of $f(t)$

$$d(w) = \frac{dw}{2\pi}\int_{-\infty}^{\infty} f(t)e^{-jwt}dt$$

  - Normalize $d(w)$ by dividing by $\frac{dw}{2\pi}$

$$\frac{d(w)}{dw/2\pi} = F(w) = \int_{-\infty}^{\infty} f(t)e^{-jwt}dt$$

  - $F(w)$ is complex and is known as the Fourier integral.

$F(w) = \mathrm{Re}[F(w)] + j\mathrm{Im}[F(w)] = |F(w)|e^{j\phi(w)}$ $(Volts/Hz)$

  - Amplitude spectral density

$|F(w)| = \sqrt{\{\mathrm{Re}[F(w)]\}^2 + \{\mathrm{Im}[F(w)]\}^2}$ (V^2/Hz^2)

  - Phase angle

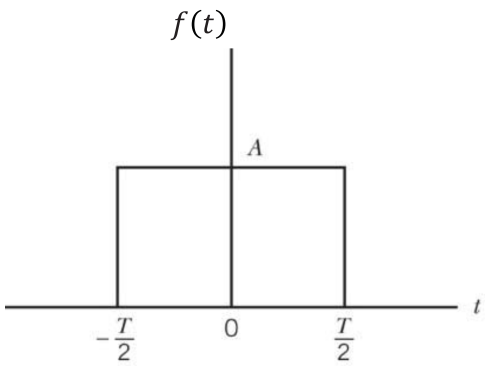
$\phi(w) = tan^{-1}[\mathrm{Im}[F(w)]/\mathrm{Re}[F(w)]]$ (degree)

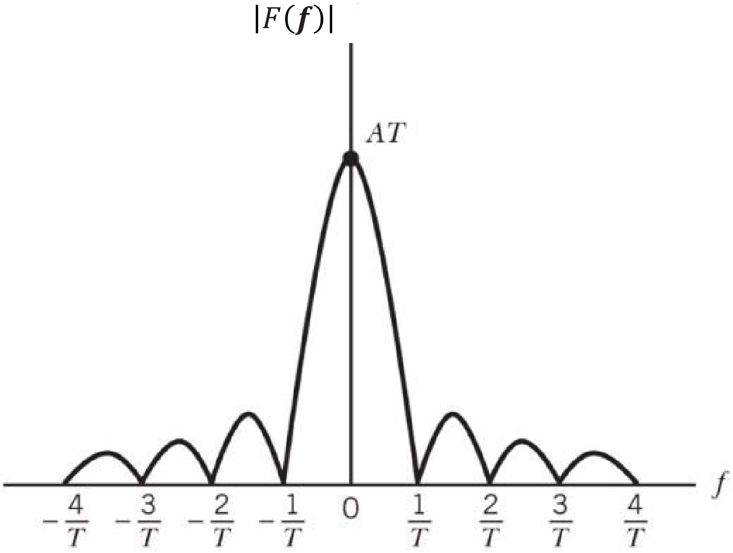
Fourier Transform (cont'd)

- Inverse Fourier transform of $F(w)$

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(w)e^{jwt}dw = \int_{-\infty}^{\infty} F(f)e^{j2\pi ft}df$$

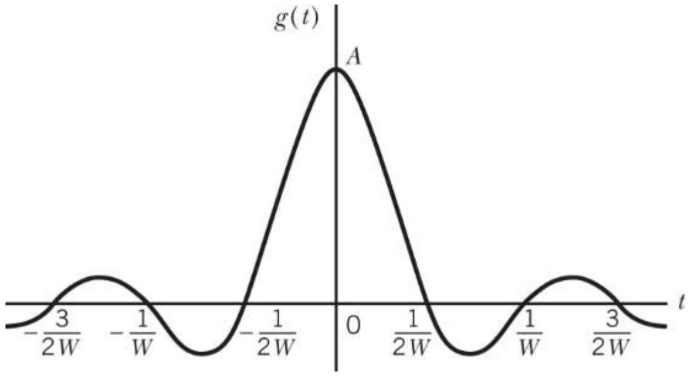
- Rectangular pulse

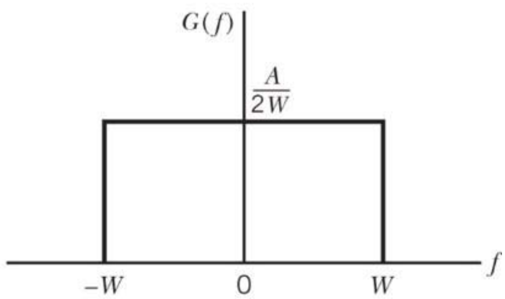




Fourier Transform (cont'd)

- sinc pulse
  - $g(t) = A \cdot \mathrm{sinc}(2Wt)$





Fourier Transform Property

Property	Mathematical Description
1. Linearity	$ag_1(t) + bg_2(t) \rightleftharpoons aG_1(f) + bG_2(f)$ where a and b are constants
2. Time scaling	$g(at) \rightleftharpoons \frac{1}{\lvert a\rvert} G\left(\frac{f}{a}\right)$ where a is a constant
3. Duality	If $g(t) \rightleftharpoons G(f)$, then $G(t) \rightleftharpoons g(-f)$
4. Time shifting	$g(t - t_0) \rightleftharpoons G(f)\exp(-j2\pi f t_0)$
5. Frequency shifting	$\exp(j2\pi f_c t)g(t) \rightleftharpoons G(f - f_c)$
6. Area under $g(t)$	$\int_{-\infty}^{\infty} g(t)\,dt = G(0)$
7. Area under $G(f)$	$g(0) = \int_{-\infty}^{\infty} G(f)\,df$
8. Differentiation in the time domain	$\frac{d}{dt}g(t) \rightleftharpoons j2\pi f G(f)$
9. Integration in the time domain	$\int_{-\infty}^{t} g(\tau)\,d\tau \rightleftharpoons \frac{1}{j2\pi f}G(f) + \frac{G(0)}{2}\delta(f)$
10. Conjugate functions	If $g(t) \rightleftharpoons G(f)$, then $g^*(t) \rightleftharpoons G^*(-f)$
11. Multiplication in the time domain	$g_1(t)g_2(t) \rightleftharpoons \int_{-\infty}^{\infty} G_1(\lambda)G_2(f-\lambda)\,d\lambda$
12. Convolution in the time domain	$\int_{-\infty}^{\infty} g_1(\tau)g_2(t-\tau)\,d\tau \rightleftharpoons G_1(f)G_2(f)$
13. Rayleigh's energy theorem	$\int_{-\infty}^{\infty} \lvert g(t)\rvert^2 dt = \int_{-\infty}^{\infty} \lvert G(f)\rvert^2 df$

Some Important FT Properties

- Linearity (Superposition)
  - Let $g_1(t) \rightleftharpoons G_1(f)$ and $g_2(t) \rightleftharpoons G_2(f)$. Both c_1 and c_2 are constants.

$$c_1 g_1(t) + c_2 g_2(t) \rightleftharpoons c_1 G_1(t) + c_2 G_2(t)$$

- Duality
  - If $g(t) \rightleftharpoons G(f)$, then

$$G(t) \rightleftharpoons g(-f)$$

  - See page 10 and 11
- Time Shifting
  - If $g(t) \rightleftharpoons G(f)$, then

$$g(t - t_0) \rightleftharpoons G(f)e^{-j2\pi f t_0}$$

  - The **amplitude** of $G(f)$ is **unaffected** by the time shift, but its **phase** is **changed** by the linear factor $\boldsymbol{-j2\pi f t_0}$.

Some Important FT Properties (cont'd)

- Frequency Shifting (***Modulation*** theorem)
  - If $g(t) \rightleftharpoons G(f)$, then
  $e^{j2\pi f_c t} g(t) \rightleftharpoons G(f - f_c)$, where f_c is a constant.
  - The **multiplication** of $g(t)$ by $\boldsymbol{e^{j2\pi f_c t}}$ is equivalent to **shifting** its Fourier transform $G(f)$ in the positive direction by $\boldsymbol{f_c}$.

Some Important FT Properties (cont'd)

- Convolution in the Time Domain
  - Let $g(t) \rightleftharpoons G(f)$ and $h(t) \rightleftharpoons H(f)$, then
  $$\int_{-\infty}^{\infty} g(\tau) \cdot h(t - \tau)\, d\tau \rightleftharpoons G(f) \cdot H(f)$$
  - The ***convolution*** of ***two signals*** in the ***time domain*** is transformed into the ***multiplication*** of their ***individual Fourier transform*** in the ***frequency domain***.
  - A signal passes through a system, and then generates an output. Obviously, it is equivalent to the convolution of the signal with the impulse response of the system.
  - It can be viewed as a **filtering** process.

Some Important FT Properties (cont'd)

- Multiplication in the Time Domain
  - Let $g_1(t) \rightleftharpoons G_1(f)$ and $g_2(t) \rightleftharpoons G_2(f)$, then

$$g_1(t)g_2(t) \rightleftharpoons \int_{-\infty}^{\infty} G_1(f) \cdot G_2(f-\lambda)\, d\lambda$$

  - The ***multiplication*** of ***two signals*** in the ***time domain*** is transformed into the ***convolution*** of their ***individual Fourier transform*** in the ***frequency domain***.

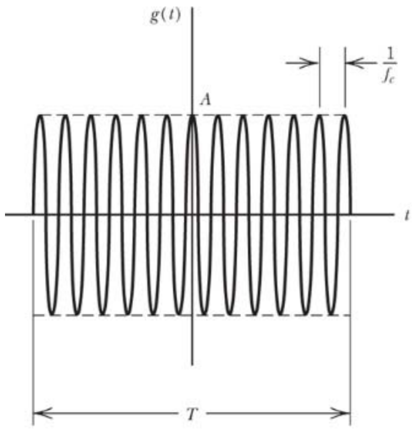
  - **Multiplication** in the time domain $\rightleftharpoons$ **modulation** in the frequency domain

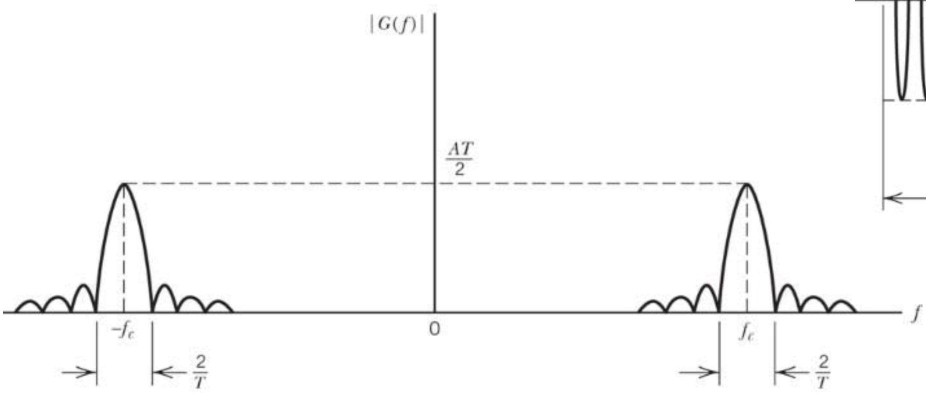
$$e^{j2\pi f_c t} g(t) \rightleftharpoons G(f - f_c)$$

$$g(t)cos(2\pi f_c t + \phi) \rightleftharpoons \frac{e^{j\phi}}{2} G(f - f_c) + \frac{e^{-j\phi}}{2} G(f + f_c)$$

Template of Modulation Scheme

- Consider the pulse signal $g(t)$, which consists of a sinusoidal wave of amplitude $\boldsymbol{A}$ and frequency $\boldsymbol{f_c}$, extending in duration from $t = -T/2$ to $t = T/2$.
  - $g(t) = A \cdot rect\left(\frac{t}{T}\right) \cdot cos(2\pi f_c t)$





Why Modulation?

- ❑ Spectrum efficiency

- ❑ Reduce antenna size
  - The length of antenna L
    $L \geq \frac{1}{10}$(the wavelength of electromagnetic transmission of analog waveform)
  - The relationship between wavelength (λ) and frequency (f)
    $\lambda = c/f$
    - $c = 3 \times 10^8$ m/sec
  - Example: The frequency of an electromagnetic wave $f = 10\ KHz$
    $$\lambda = \frac{(3\times10^8\ m/s)}{10^4/s} = 3\times10^8\ m$$
  - The length of antenna $\geq 0.1\lambda$ =3 Km
  - WIFI in 2.4 GHz and 5 GHz, $\lambda = ?$

Random Process

- ❑ **Statistical** Averages
  - The **expected value** or **mean** of a random variable $\boldsymbol{X}$ is defined by
    $$\boldsymbol{\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx}$$
  - $\boldsymbol{E}[\cdot]$: the statistical expectation operator
  - $\boldsymbol{f_X(x)}$: Probability density function (PDF) of the random variable $\boldsymbol{X}$
  - The mean $\boldsymbol{\mu_X}$ locates the center of gravity of the area under the probability density curve of the random variable $\boldsymbol{X}$.

Random Process (cont'd)

- Let $\boldsymbol{X}$ is a ransom variable, and let $\boldsymbol{g(X)}$ is a real-valued function.
  - Let $\boldsymbol{Y = g(X)}$, $\boldsymbol{Y}$ is also a random variable.
  - Then, the expected value of $\boldsymbol{Y}$ can be found, based on the PDF $\boldsymbol{f_Y(y)}$, as

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y)\, dy \;\; or \;\; E[g(X)] = \int_{-\infty}^{\infty} g(X) f_X(x)\, dx$$

  - It can be viewed as generalizing the concept of expected value to an arbitrary function $\boldsymbol{g(X)}$ of a random variable of $\boldsymbol{X}$.

Random Process (cont'd)

- Second central moment
  - The *variance* of the random variable $\boldsymbol{X}$

$$var[X] = E\left[(X-\mu_X)^2\right] = \int_{-\infty}^{\infty} (x-\mu_X)^2 f_X(x)\, dx$$

  - $\boldsymbol{var[X] = \sigma_X^2}$, constraining the effective width of the pdf $\boldsymbol{f_X(x)}$ of the random variable $\boldsymbol{X}$ about the mean $\boldsymbol{\mu_X}$
  - $\boldsymbol{\sigma_X}$: the **standard deviation** of the random variable $\boldsymbol{X}$
- **Autocorrelation** and **autocovariance** function of the random variable $\boldsymbol{X}$
  - Autocorrelation: $R_X(t,s) = E[X(t)X(s)] = R_X(s-t) = R_X(\tau)$
  - Autocovariance: $C_X(t,s) = E[(X(t)-\mu_X)(X(s)-\mu_X)] = R_X(s-t) - \mu_X^2$
  - $\tau = s - t$

Classification of Random Process

- Strict-sense stationary (SSS)
  - The statistical properties of a continuous-time random process $\boldsymbol{X(t)}$ are **invariant** under an arbitrary time shift.
  - Stationary to **order *N***
  - Mean

$\boldsymbol{E[X(t)] = E[X(t - t_0)] = \mu_X}$ for all $\boldsymbol{t}$ (time-invariant)

  - Variance

$\boldsymbol{var[X(t)] = \sigma_X^2}$

Classification of Random Process (cont'd)

- ***Wide-sense stationary*** (WSS)
  - Stationary to **order two**
  - A random process $\boldsymbol{X(t)}$ satisfies the conditions
    - $\boldsymbol{\mu_X(t) = \mu_X = constant}$
    - $\boldsymbol{R_X(t_1, t_2) = R_X(t_2 - t_1)}$.
  - All strictly stationary processes are wide-sense stationary, but no all wide-sense stationary processes are strictly stationary.

Random Process (cont'd)

- Cyclostationary process
  - A continuous-time random process $X(t)$ is strict-sense cyclostationary with period T if its statistical properties are invariant under a time shift of kT.
  - A process $X(t)$ is wide-sense cyclostationary with period T if its mean and autocorrelation function is periodic in time with period T.
  - A wide-sense cyclostationary process can be made WSS by

$\boldsymbol{X(t) = X(t-\tau)}$, where $\boldsymbol{\tau}$ is uniformly distributed over $[\mathbf{0}, \boldsymbol{T}]$..

Time Average and Ergodic Process

- ***Ensemble*** averages
  - The mean of a stochastic process at some fixed time $\boldsymbol{t_k}$ is the expectation of the random variable $\boldsymbol{X(t_k)}$ that describes ***all possible values*** of the sample functions of the process observed at time $\boldsymbol{t = t_k}$.

- ***Ergodicity***
  - It is more convenient to observe a single sample function for a long period of time.
  - For a sample function $\boldsymbol{x(t)}$, the time average of the mean value over an observation period $\mathbf{2}\boldsymbol{T}$ is

$$\mu_{x,T} = \frac{1}{2T}\int_{-T}^{T} x(t)\, dt$$

Ergodic Process (cont'd)

- For many stochastic processes of interest in communications, the ***time averages*** and ***ensemble averages*** are ***equal***, a property known as ***ergodicity***.
- The property implies that whenever ***an ensemble average*** is ***required***, we may ***estimate*** it by using ***a time average***.

Power Spectral Density

- **Power spectral density (PSD)** $S_X(f)$ of the random process $X(t)$
  - The Fourier transform of the the auto-correlation function $R_X(\tau)$

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau$$

and

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df$$

  - Basic relations in the theory of spectral analysis of random process
  - As called the Einstein-Wiener-Khintchine relations
    - If either the auto-correlation function or power spectral density of a random process is known, the other can be found.

Power Spectral Density (cont'd)

- Properties of the PSD for a WSS random process
  - $S_X(f) \geq 0$
    - The PSD is always nonnegative.
  - $S_X(0) = \int_{-\infty}^{\infty} R_X(\tau)d\tau$
    - The zero-frequency value of the PSD is equivalent to the total area under the graph of the autocorrelation function.
  - $E[X^2(t)] = \int_{-\infty}^{\infty} S_X(f)df$
    - The mean-square value equals the total area under the graph of the PSD.
  - $S_X(-f) = S_X(f)$
    - The PSD of a **real-valued** random process is an **even function** of **frequency**.

Power Spectral Density (cont'd)

- Relation between the PSD of the input and output random process

$$S_Y(f) = H(f)H^*(f)S_X(f)$$

$$S_Y(f) = |H(f)|^2 S_X(f)$$

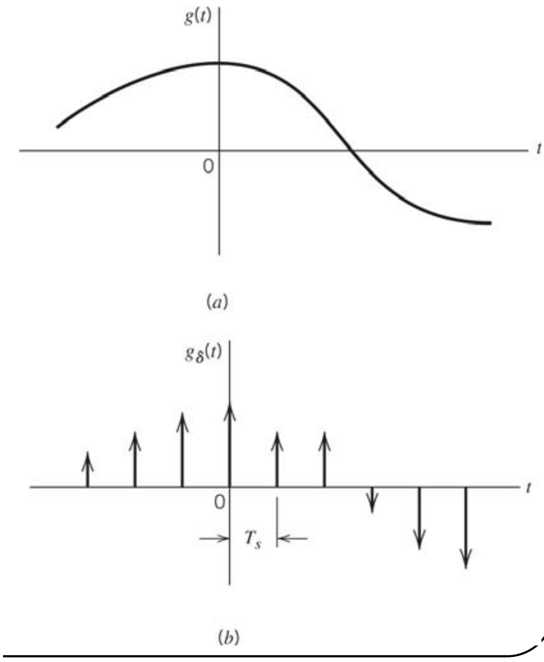
  - $S_Y(f)$: The PSD of the output process $Y(t)$
  - $S_X(f)$: The PSD of the output process $X(t)$
  - $H(f)$: The transfer function $H(f)$ of a linear filter (system)

  - The PSD of the output process $Y(t)$ equals the PSD of the input process $X(t)$ multiplied by the squared magnitude of the transfer function $H(f)$ of the filter (system).

Sampling Theory

- An analog signal $g(t)$ is sampled by a frequency f_s (uniformly sample), that is equivalent to $1/T_s$.
  - f_s: the sampling rate
  - T_s: the sampling period
  - Ideal sampled signal $g_\delta(t)$

$$g_\delta(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\,\delta(t - nT_s)$$



Sampling Theory (cont'd)

- Fourier transform of $g_\delta(t)$

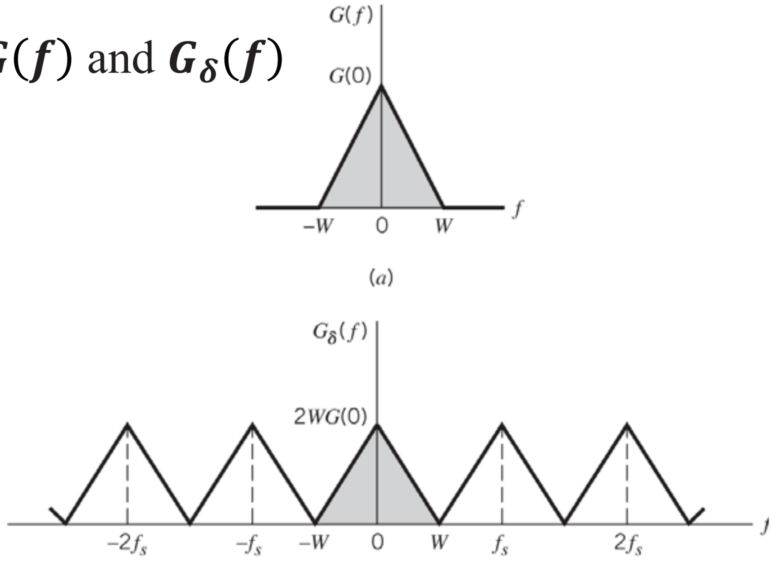
$$g_\delta(t) \rightleftharpoons f_s \sum_{m=-\infty}^{\infty} G(f - mf_s)$$

  - The process of uniformly sampling an analog (contnuous-time) signal of finite energy results in a periodic spectrum with a period equal to the sampling rate.

- Consider a strictly band-limited signal $g(t)$ with a finite energy and its $G(f) = 0$ for $|f| \geq W$
  - Choose the sampling period $T_s = 1/2W$
  - $f_s = 2W$

Sampling Theory (cont'd)

❑ Spectra of $\boldsymbol{G(f)}$ and $\boldsymbol{G_\delta(f)}$

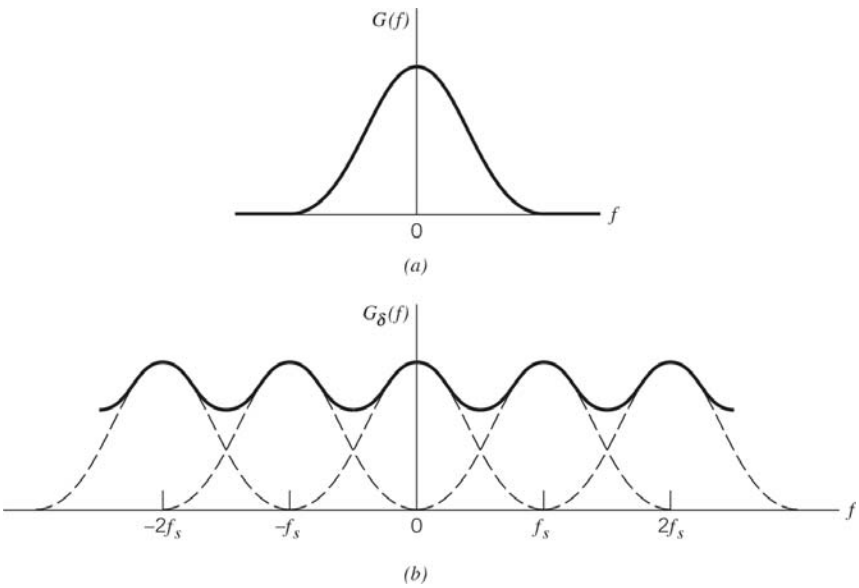


■ The signal $\boldsymbol{g(t)}$ may be completely recovered from a knowledge of its samples taken at the rate of 2W samples per-second.

- Sampling rate $\boldsymbol{f_s \geq 2W}$ (Nyquist rate: $\boldsymbol{f_s} = 2\mathrm{W}$)

Sampling Theory (cont'd)

❑ Aliasing Spectrum of $\boldsymbol{G_\delta(f)}$ with $\boldsymbol{f_s < 2W}$



Discrete Fourier Transform (DFT)

- The signal involved in DFT/FFT have to satisfy the sampling theory.

$$f_s \geq 2f_{max}$$

  - f_s: the sampling frequency
  - f_{max}: the maximum frequency component of the sampled signal
- The samples are uniformly spaced by T_s seconds, the spectrum of the signal becomes periodic, repeating every $f_s = 1/T_s$ Hz.
- Let N denote the number of ***frequency samples*** contained in a interval f_s.

DFT (cont'd)

- Hence, the ***frequency resolution*** involved in the numerical computation of the Fourier transform is defined by

$$\Delta f = \frac{f_s}{N} = \frac{1}{NT_s} = \frac{1}{T}$$

  - T: the ***total duration*** of the signal ($T = NT_s$)

DFT (cont'd)

- The DFT of x_n is defined as

$$X_k = X_{k\Omega} = F[x_n] = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi}{N}kn}, \qquad k = 0, 1, \dots, N-1$$

  - k: the frequency index
  - $\{X_0, X_1, \dots, X_{N-1}\}$: Transform sequences
  - N -point DFT processing
  - $F[\cdot]$: DFT operation
  - $n = nT_s$ ($\because t = nT_s$)
  - Ω: the discrete time frequency
  - $k = k\Omega = k2\pi f = k2\pi/T = 2\pi k/NT_s$ ($\because w = k\Omega, T = NT_s$)
- Refer to Fourier transform
  - $F(w) = \int_{-\infty}^{\infty} f(t)e^{-jwt}dt$

DFT (cont'd)

- The IDFT of X_k is given as

$$x_n = \frac{1}{N}\sum_{k=0}^{N-1} X_k e^{-j\frac{2\pi}{N}kn}, \qquad n = 0, 1, \dots, N-1$$

- Transform pair: DFT vs. IDFT
  - The DFT of a data sequence (in the time domain) of length ***N*** as an ***N*-point** DFT
    - Time domain **samples**
  - The IDFT of a data sequence (in the frequency domain) of length ***N*** as an ***N*-point** IDFT
    - Frequency domain **spectra (or harmonic)**

DFT (cont'd)

- The kth harmonic (spectrum) is given as
  - $X_k = \Re[X_k] + j\Im[X_k]$
- The amplitude and phase of X_k can be found as
  - $|X_k| = \sqrt{\Re[X_k]^2 + \Im[X_k]^2}$
  - $\phi_k = tan^{-1}[\Im[X_k]/\Re[X_k]]$

DFT (cont'd)

- Spectrum
  - Amplitude spectrum is symmetry about $N/2$.
  - Phase function exhibits odd anti-symmetry about harmonic $N/2$.
  - $2f_{max}t = N$, so the 1st harmonic frequency $= 1/t = 2f_{max}/N$
  - The symmetry at harmonic $N/2$ is at the frequency $(N/2)(2f_{max}/N) = f_{max}$.
  - f_{max} is also called the folding frequency.